

Dart Fundamentals - Assignment 1: Library System

A simple and beginner-friendly console-based Library Management System written in Dart. This project demonstrates core Dart concepts including Object-Oriented Programming (OOP), collections, control flow, functions, and classes.

Features

- Book Modeling: Represents books with title, author, and availability status.
- Library Catalog: Add and store multiple books using Dart `List`.
- View Inventory: Display all books in the catalog along with their real-time availability.
- Borrow System: Allows borrowing available books and updating their status automatically.

Concepts Covered

Concept	Implementation in Code
Classes & Objects	`Book` and `Library` classes
Constructors	Parameterized constructors (`Book(this.title, this.author)`)
Collections	`List<Book>` to manage inventory
Control Flow & Loops	`for-in` loops to traverse books and `if` conditions to check availability
Methods	`addBook()`, `showBooks()`, `borrowBook()`, and `displayBook()`

Project Structure

```
``text
dart_fundamentals/
├── Assignment1/
│   └── library_system.dart
...

```

How to Run

Ensure you have the [Dart SDK](<https://dart.dev/get-dart>) installed on your system.

1. Navigate to the project directory:

```
``bash
cd Assignment1
...

```

2. Run the application:

```
``bash
dart library_system.dart
...

```

—

 Sample Output

```
```text
--- Central Library ---
The Three-Body Problem by Cixin Liu
Available: true
Blindsight by Peter Watts
Available: true

You borrowed Blindsight

--- Central Library ---
The Three-Body Problem by Cixin Liu
Available: true
Blindsight by Peter Watts
Available: false
```
```